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Multiphase flow simulations of a moving fluidized bed regenerator in a carbon capture unit

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ABSTRACT

To accelerate the commercialization and deployment of carbon capture technologies, computational fluid dynamics (CFD)-based tools may be used to model and analyze the performance of carbon capture devices. This work presents multiphase CFD flow simulations for the regenerator—a device responsible for extracting CO₂ from CO₂-loaded particles before the sorbent is recycled. The use of solid particle sorbents in this design is a departure from previously reported systems, where aqueous sorbents are employed. Another new feature is the inclusion of a series of perforated plates along the regenerator height. The influence of these plates on sorbent distribution is examined for varying sorbent holdup, fluidizing gas velocity, and particle size. The residence time distribution of sorbents is also measured to classify the flow regime as plug flow or well-mixed flow. The purpose of this work is to better understand the sorbent flow characteristics before reaction kinetics of CO₂ desorption can be implemented.

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1. Introduction

Combustion of fossil fuels in thermal power plants is one of the largest sources of greenhouse gas emissions. In addition to promoting energy efficiency and the use of renewable energy sources, carbon capture and sequestration have also been investigated as an alternate approach to limit carbon dioxide (CO₂) emissions. Post-combustion CO₂ capture using amine-based sorbents is considered as a candidate approach due to the technology maturity of amine-based sorbents [1–3]. However, at present, only a few pilot-scale carbon capture plants are functional, and commercial scale projects are yet to be constructed (see Wang et al. [4] and the database maintained by Carbon Capture and Sequestration Technologies MIT [5]). To accelerate development and commercialization of carbon capture technologies, computational fluid dynamics (CFD)-based modeling tools can be used to simulate and optimize the carbon capture systems. To this end, the Carbon Capture Simulation Initiative (CCSI) was established as a collaborative effort between national laboratories, universities, and industry partners in the United States to develop and deploy computational tools to promote rapid commercialization of carbon capture technologies. This work, performed as part of CCSI's efforts, presents a multiphase CFD model for simulating the flow dynamics in the regenerator unit of a conceptual sorbent-based carbon capture system.

The performance, reliability, and operating cost of the proposed regenerator unit are currently being assessed (for details, refer to Lee et al. [6]). Flow behavior in the regenerator is expected to strongly influence the CO₂ release characteristics of the sorbent particles, thereby affecting the overall efficiency of the carbon capture unit. At this stage, the effects of operating conditions and sorbent properties on flow dynamics in the regenerator are not properly understood. This work investigates the influence of operating conditions, such as sorbent mass holdup, gas velocity, and sorbent particle size on bed flow behavior in a regenerator design with a series of horizontal perforated plates. Because the regenerator is associated with the bulk of the energy penalty of a carbon capture unit [3,7], optimization of the regeneration stage is the critical first step toward developing commercially viable systems. Characterization of the flow dynamics in a regenerator is the precursory step before detailed and computationally more expensive chemical reaction kinetics models can be implemented.

2. Background

A number of recent works have presented post-combustion carbon capture systems based on chemical absorption/adsorption of CO₂ from exhaust flue gases [1–4,7–11]. A typical plant layout is presented in Fig. 1, although the design of individual devices and the sorbent type may vary. All carbon capture systems consist of an adsorber and a regenerator. In the adsorber (also called 'absorber'), exhaust flue gases from a power plant react with solid or aqueous chemical sorbents capable of extracting CO₂ from the surrounding gas phase. Details regarding solid particulate sorbents can be found in Lee et al. [12] and Mebane

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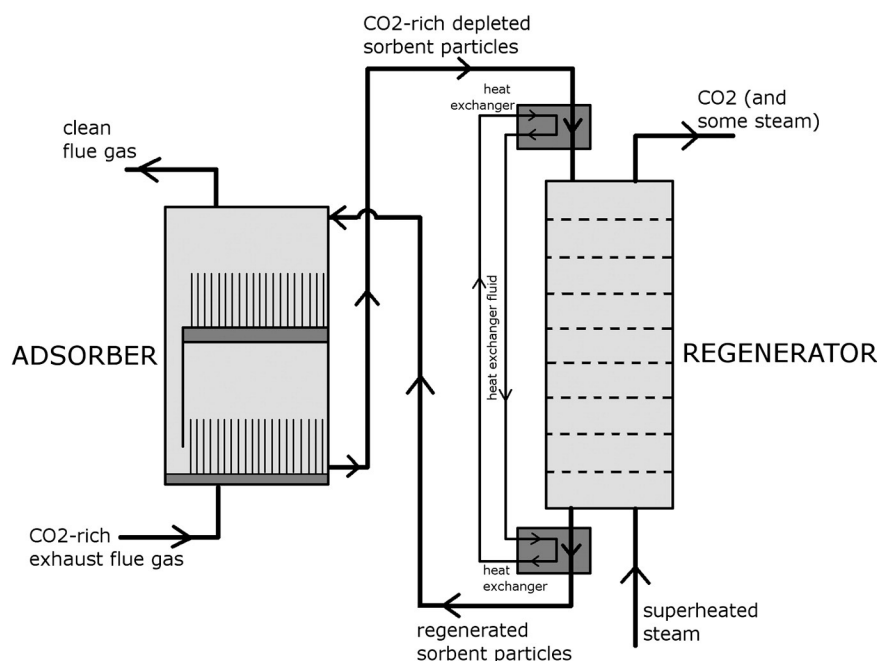


Fig. 1. Simplified schematic of the carbon capture system design proposed by the Carbon Capture Simulation Initiative (CCSI) team.

et al. [13]. After additional processing steps, the clean exhaust gases subsequently are released into the atmosphere. Meanwhile, the loaded CO_2 -rich sorbent is transferred to the regenerator. The regenerator (also called the 'stripper' or 'desorber') is responsible for extracting the adsorbed CO_2 from the sorbent through the reverse chemical reaction. The regenerated sorbent is then recycled back to the adsorber for reuse. Typically, the regenerator operates at a higher temperature than the adsorber as the CO_2 -release rate is larger at higher temperatures, whereas lower temperatures are more favorable for CO_2 capture. A heat exchange between outlet streams from the regenerator and the adsorber improves the efficiency of the capture unit.

Efforts to model the dynamic behavior of a carbon capture unit have focused on modeling the absorber and regenerator individually [1,2,8,11,14], as well as with both devices together as a complete cycle [7,9,10]. In these works, 1D dynamics models of the devices are developed based on macroscopic conservation equations supplemented with constitutive relationships available in the literature. The sorbents used are typically aqueous solutions of monoethanolamine-based compounds. The microscopic flow behavior of the aqueous sorbent in these devices is not studied and transverse variations are ignored. Often, a plug flow behavior is considered along the device height, an assumption whose validity in systems with solid sorbents, as will be shown, is questionable. The flow dynamics of sorbent particles in such fluidized bed carbon capture systems needs to be studied in greater detail.

There have been a number of experimental studies that investigate the use of solid sorbents for CO_2 -capture. A number of works characterize the performance of inorganic sorbents as well as newer amine-based (see, for example, Abanades et al. [15], Sjostrom and Krutka [16], Gray et al. [17]). Very few studies model the flow behavior of sorbents inside the carbon capture devices. Chalermisinsuwan et al. [18] investigate the flow dynamics and reaction kinetics of an inorganic sorbent inside a closed loop capture system using CFD. They show that the capture efficiency may be improved by reducing the flow rate of CO_2 into the device, or by using a better sorbent. Measurements of the granular diffusion coefficients demonstrate significant axial and radial mixing of sorbents inside the devices, which limits the use of the axial plug flow model for such solid sorbent-based systems. Thus, it is important to characterize the axial mixing of sorbents in the capture devices

(as discussed in Section 4.2 of this work). A smaller-scale capture unit studied in Khongprom and Gidaspow [19], similar in design as the system in Chalermisinsuwan et al. [18], was reported to have a better capture efficiency. The improvement was attributed to better sorbent distribution and flow in the more compact adsorber.

As CFD-based studies specific to capture systems with solid sorbents are scarce, the existing literature on fluidized bed reactors may provide better understanding of the flow in these carbon capture devices. The operating flow regime strongly governs the performance of fluidized bed reactors. The fluidization regime is governed by the upward velocity of gases in the bed, often expressed as a multiple of the superficial minimum fluidization velocity $v_{g,mf}$. At minimum fluidization, the gas pressure drop is balanced by the weight of the bed. The minimum fluidization velocity $v_{g,mf}$ is given by:

$$v_{g,mf} = \frac{(\rho_s - \rho_g) g d_p^2 (1 - \phi_{s,mf})^3}{150 \mu_g \phi_{s,mf}}, \quad (1)$$

where ρ_s and ρ_g are densities of the solid particles and fluidizing gas, respectively; d_p is the particle diameter; μ_g is gas viscosity; and g is acceleration due to gravity. The bed solid volume fraction at minimum fluidization $\phi_{s,mf}$ is assumed to have a constant value of 0.55 in this work. For beds operating in the bubbling fluidization regime, superficial gas velocities on the order of 1 to 10 times $v_{g,mf}$ are typical. For the current regenerator, a downward axial flow corresponding to the flow rate of sorbents is superimposed on the bubbling bed, and the design is called the "moving bed regenerator" [20,21].

The mean solid volume fraction and distribution of solid particles are likely to influence the performance of the moving bed regenerator. For relating the mean solid fraction with the gas velocity and particle properties, Jamialahmadi et al. [22] suggested using the Kozeny–Carman equation or the Ergun equation. The Ergun equation [23] is expressed as:

$$\frac{\Delta P}{\Delta y} = \frac{150 \mu_g v_{g,sup}}{d_p^2} \frac{\phi_s^2}{(1 - \phi_s)^2} + \frac{1.75 \rho_g v_{g,sup}^2}{d_p} \frac{\phi_s}{(1 - \phi_s)^3}, \quad (2)$$

where $\Delta P/\Delta y$ is the pressure drop per unit height and $v_{g,\text{sup}}$ is the superficial gas. For fluidized beds operating beyond minimum fluidization, the pressure drop per unit height $\Delta P/\Delta y$ is balanced by gravity and buoyancy forces and can be replaced by $(\rho_s - \rho_g)\phi_s g$. Jamialahmadi et al. [22] showed good agreement between the Ergun equation and experimentally measured pressure drop in a porous bed. To account for the downward velocities of particles in a moving bed, Yoon and Kunii [21] suggested that superficial gas velocity $v_{g,\text{sup}}$ in Eq. (2) be replaced by the product of bed voidage and relative velocity between the gas and solids.

Although the mean solid fraction in fluidized columns is predicted reasonably well using the Ergun equation, no information regarding the local bed microstructure is obtained. Local inhomogeneities in the form of dynamic particle clusters and gas bubbles are seen from instantaneous snapshots of a fluidized bed (for examples, refer to Zimmermann and Taghipour [24] and Taghipour et al. [25]). In a series of snapshots, McKeen and Pugsley [26] show that flow behavior is dynamic, and particle clusters coalesce and disintegrate over time scales shorter than a few seconds. Agrawal et al. [27] and Holloway and Sundaresan [28] have shown that these particle clusters, in turn, strongly affect the macroscopic flow dynamics and reaction kinetics, respectively. Detailed CFD simulations which capture this dynamic behavior are needed to correctly characterize flow in the regenerator.

Operating conditions, such as gas velocity, and material properties, such as particle size, are expected to have a strong influence on the flow regime, and hence, the distribution of sorbent particles in the regenerator. Much of the existing literature on fluidized bed columns studies systems without any internal geometric obstructions. The addition of a series of horizontal perforated plates in the regenerator can potentially result in significant differences in flow. To determine a suitable operating regime in the regenerator, the influence of sorbent holdup, steam velocity, and particle size on flow dynamics is investigated. This work focuses on characterizing the flow dynamics of sorbent particles, and the chemical kinetics of CO_2 desorption are not included. In a study by Li and Guenther [29], the inclusion of chemical kinetics describing interconversion between ozone and oxygen had a noticeable but relatively small effect on particle and gas flow dynamics. Although the flow dynamics in the regenerator may change when the chemical reactions are considered, an understanding of the flow inside the regenerator for an isothermal, non-reactive system is a reasonable first step toward development of a detailed model capable of accurately predicting regenerator behavior.

3. Model

Regenerator CFD simulations were performed using the multiphase two-fluid model. Details about the two-fluid model can be found in Gidaspow [30], and information regarding the implementation in the open-source code MFIX can be found in Syamlal et al. [31] and Syamlal [32]. A summary of the two-fluid model equations used in this work is presented in Table 1.

The 2D regenerator model schematic is presented in Fig. 2. The height of the CCSI regenerator is 10 m, typical of commercial-scale devices. A transverse regenerator width of 2 m is simulated. As the model represents a central section of the full-scale device, free-slip side walls are used to simulate the symmetry conditions in the transverse direction. The influence of other side wall boundary conditions is discussed in Section 3.1.

The solid particles enter the regenerator through a sorbent inlet at the top of the device and leave through a sorbent outlet at the base. Superheated steam, used to raise the temperature of the loaded sorbent, is injected through gas inlets located on the sides at the base of the regenerator. The corresponding outlets for gas, modeled as pressure outflow boundaries, are located at the top of the device on either side of the sorbent inlet. The gas pressure at the top gas outlet boundaries is set to the standard atmospheric pressure. The sorbent particles and steam entering

Table 1
Summary of two-fluid model equations.

<i>Mass conservation</i>	
$\frac{\partial(\rho_s \phi_s)}{\partial t} + \nabla \cdot (\rho_s \phi_s \mathbf{v}_s) = 0$	(3)
$\frac{\partial(\rho_g \phi_g)}{\partial t} + \nabla \cdot (\rho_g \phi_g \mathbf{v}_g) = 0$	(4)
<i>Momentum conservation</i>	
$\frac{\partial(\rho_s \phi_s \mathbf{v}_s)}{\partial t} + \nabla \cdot (\rho_s \phi_s \mathbf{v}_s \mathbf{v}_s) = -\nabla \cdot \sigma_s - \phi_s \nabla \cdot \sigma_g + f_{gs} + \rho_s \phi_s \mathbf{g}$	(5)
$\frac{\partial(\rho_g \phi_g \mathbf{v}_g)}{\partial t} + \nabla \cdot (\rho_g \phi_g \mathbf{v}_g \mathbf{v}_g) = -\phi_g \nabla \cdot \sigma_g - f_{gs} + \rho_g \phi_g \mathbf{g}$	(6)
<i>Granular energy conservation (algebraic formulation adapted from [33])</i>	
$\theta_s = \left[\frac{-K_1 \phi_s \text{tr}(\mathbf{D}_s) + \sqrt{[K_1 \text{tr}(\mathbf{D}_s) \phi_s]^2 + 4K_4 \phi_s [K_2 \text{tr}^2(\mathbf{D}_s) + 2K_3 (\mathbf{D}_s : \mathbf{D}_s)]}}{2\phi_s K_4} \right]^2$	(7)
where $\mathbf{D}_s = \frac{1}{2}[\nabla \mathbf{v}_s + (\nabla \mathbf{v}_s)^T]$, $K_1 = 2(1 + e_{pp})\rho_s g_0$, $K_2 = \frac{4d_p \rho_s (1 + e_{pp}) \phi_s g_0}{3\sqrt{\pi}} - \frac{2}{3}K_3$, $K_3 = \frac{d_p \rho_s}{2} \left\{ \frac{\sqrt{\pi}}{3(3 - e_{pp})} [0.5(1 + 3e_{pp}) + 0.4(1 + e_{pp})(3e_{pp} - 1)\phi_s g_0] + \frac{8d_p g_0 (1 + e_{pp})}{5\sqrt{\pi}} \right\}$, $K_4 = \frac{12(1 - e_{pp}^2)\rho_s g_0}{d_p \sqrt{\pi}}$	
<i>Gas phase stress</i>	
$\sigma_g = pI - \mu_g [\nabla \mathbf{v}_g + (\nabla \mathbf{v}_g)^T] - \frac{2}{3}(\nabla \cdot \mathbf{v}_g)I$	(8)
<i>Solid phase stress</i>	
$\sigma_s = [\rho_s \phi_s (1 + 4\eta \phi_s g_0) \theta_s - \eta \mu_b (\nabla \cdot \mathbf{v}_s)]I - 2\mu_s S$ where $S = \frac{1}{2}[\nabla \mathbf{v}_s + (\nabla \mathbf{v}_s)^T] - \frac{1}{3}(\nabla \cdot \mathbf{v}_s)I$, $\mu_s = \left(\frac{2}{3}\alpha\right) \left[\frac{\mu^*}{(g_0 \eta^2 - 1)} (1 + \frac{8}{3}\phi_s \eta g_0) (1 + \frac{8}{3}\eta(3\eta - 2)\phi_s g_0) + \frac{2}{3}\eta \mu_b \right]$, $\mu_b = \frac{256\eta d_p^2 g_0}{5\pi}$, $\mu^* = \frac{\mu}{1 + (\rho_s \phi_s)^2 g_0 \eta}$, $\mu = \frac{5\rho_s d_p \sqrt{\pi} \theta_b}{96}$, $\eta = \frac{1 + e_{pp}}{2}$, $\alpha = 1.6$	(9)
<i>Gas-solid drag (Wen and Yu [34])</i>	
$f_{gs} = \beta_{gs}(\mathbf{v}_g - \mathbf{v}_s)$, $\beta_{gs} = \frac{3}{4}C_D \frac{\rho_g (1 - \phi_s) \phi_s \mathbf{v}_g - \mathbf{v}_s }{d_p} (1 - \phi_s)^{-2.65}$ where $C_D = \begin{cases} \frac{24}{Re_p} (1 + 0.15 Re_p^{0.687}), & Re_p < 1000 \\ 0.44, & Re_p \geq 1000 \end{cases}$	(10)
$Re_p = \frac{(1 - \phi_s) \rho_g d_p \mathbf{v}_g - \mathbf{v}_s }{\mu_g}$	

the regenerator must pass through a series of horizontal perforated plates before exiting. An infinitesimally thin semipermeable layer adjacent to the top gas outlets prevents the escape of sorbents through these steam exits.

The inlet flow rate of solids at the top sorbent boundary is set to a constant value of 6 kg/s, which corresponds to the designed flow rate of approximately 500 tons/h in the full-scale device. The sorbent outlet at the bottom of the regenerator is modeled as a mass outflow boundary. The flow rate of solids at the bottom sorbent outlet is identical to the solids flow rate at the top sorbent inlet. The gas flow rates at the sorbent inlet and outlet are set to 7.9×10^{-3} kg/s to ensure that the gas and solids velocities remain the same at these boundaries.

In an actual regenerator, CO_2 is released into the gas phase as sorbents are refreshed. In this work, reactions are not modeled, and all gases in the system, including CO_2 , are collectively represented by the superheated steam. The gas flow rate is varied systematically to span the entire range of gas velocities expected in the regenerator. The release of CO_2 may result in a variation of the gas flow rate with height, and this effect will be accounted for in a future work.

The solid and gas phase properties used in the simulations are summarized in Table 2. Considering that the sorbent is comprised of porous silica particles impregnated with polyethylenimine, a relatively small particle–particle restitution coefficient value of 0.5 was used. It is expected that inelastic energy dissipation will be more for these highly porous and coated sorbent particles. McKeen and Pugsley [26] and Zimmermann and Taghipour [24] show that the restitution coefficient, varied between 0.50 and 0.99, does not have a significant influence on bed behavior, particularly at smaller gas velocities. For the range of gas velocities considered in this work ($1\text{--}10v_{g,\text{mf}}$), the results are not expected to change significantly even if the restitution coefficient

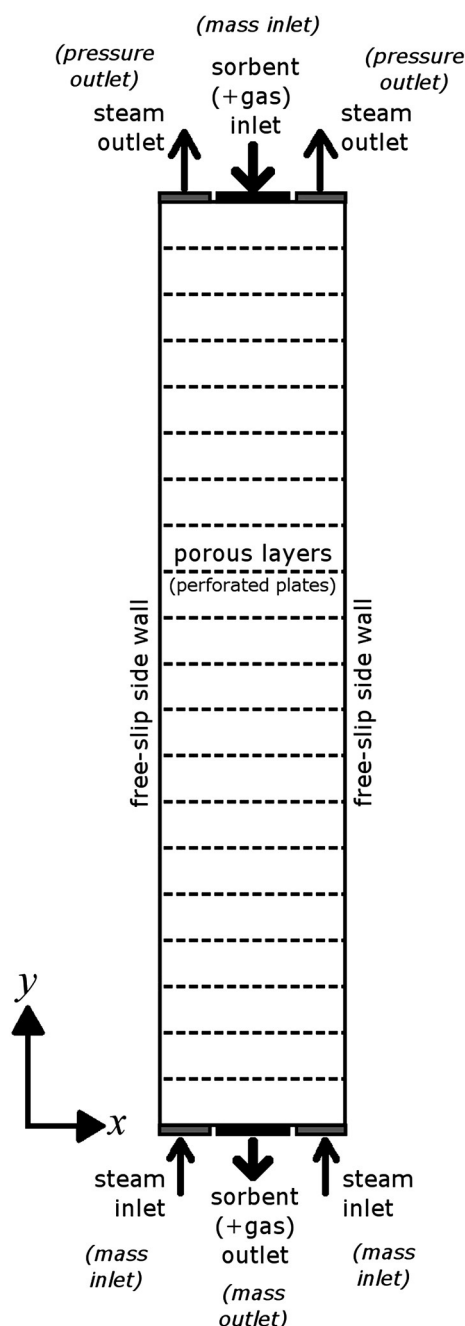


Fig. 2. Model of the regenerator simulated using the open-source code MFIX. The 2D model has a height and width of 10 m and 2 m, respectively, with layers of stationary porous media representing the perforated plates placed 0.5 m apart.

is underestimated. The actual restitution coefficient is yet unknown due to the difficulties associated with making such particle-level measurements. The particle–wall restitution coefficient was set to 0.4. The frictional properties of the particles are assigned values typical for sand (silica) as material properties for the sorbents have not yet been determined experimentally.

A number of works on fluidized bed systems have used the Wen and Yu [34] drag model [25,28,35–37]. A comparison of the several drag laws by Taghipour et al. [25] shows that no significant difference in fluidization behavior was observed using the different models. As no drag model is reported to perform significantly better than the others, the Wen and Yu [34] relationship, commonly used in many similar studies, is chosen to model the interphase drag in this work.

Table 2

Invariant parameters used in the simulations.

Property	Symbol	Value	Units
<i>Simulation parameters</i>			
Gravity	g	9.81	m/s^2
<i>Sorbent particle properties</i>			
Particle density	ρ_s	441	kg/m^3
Particle–particle restitution coefficient	e_{pp}	0.5	–
Particle–wall restitution coefficient	e_{pw}	0.4	–
Angle of internal friction	θ_{fs}	40°	–
Particle–wall friction coefficient	μ_{pw}	0.7	–
Maximum solid fraction at packing	$\phi_{s,\max}$	0.60	–
<i>Gas (steam) properties</i>			
Gas density	ρ_g	0.581	kg/m^3
Gas viscosity	μ_g	1.26×10^{-5}	–
<i>Porous plate properties</i>			
Plate porosity	ϕ_{plate}	0.80	–
Porous media particle diameter	$d_{p,\text{plate}}$	2.0	mm

A comparison of the Schaeffer [38] and Srivastava and Sundaresan [39] models by Reuge et al. [40] shows that predictions of bed expansions are qualitatively identical and differ only by a small amount quantitatively. For the purpose of predicting solid fraction profile in the bed, the choice of solids stress model is not expected to be crucial as Reuge et al. [40] show both models to yield reasonably similar results. Hence, the solids stress formulation given by Schaeffer [38] is used for closure of the solids momentum equation. Details regarding the implementation of the Schaeffer [38] model can be found in Syamlal et al. [31]. The second-order Superbee scheme is used while solving the conservation equations using MFIX.

A number of horizontal, uniformly-spaced perforated plates were placed along the height of the regenerator to help regulate particle flow and support the particle weight. In the actual device, the diameter of the perforations is expected to be a few millimeters to a few centimeters. Full-scale simulations with the perforation holes explicitly resolved would be computationally prohibitive as $O(10^6)$ cells are required for a 2D simulation. Hence, the perforated plates are approximated as stationary, uniform porous media layers that limit the volume available to sorbent particles flowing through each plate. Currently, the porosity of the stationary plate layers is set to 80%. Each perforated plate has a thickness of 5 cm, which is expected to sufficiently support the weight of sorbents between two plates. The properties of the stationary porous medium are summarized in Table 2.

All results presented in this work were obtained from flow simulations of duration 200 s or longer. The initial transients typically last for 30–70 s, depending on the operating conditions. To confirm that measurements are performed after the system attains a statistically steady state, the initial 100 s of the flow is not used to collect any statistics.

The authors note that systematic validation studies are needed to gain confidence on the accuracy of the regenerator CFD predictions, and a number of previous works have compared model predictions with experimental results (see Taghipour et al. [25], Reuge et al. [40], Lindborg et al. [41]). Some of these studies (for example, Reuge et al. [40]) address validation of the open-source code MFIX [31,32], which is also used in this study. Validation of the regenerator CFD results presented in this work is currently not possible because the CCSI pilot-scale regenerator has yet to be constructed. Grace and Taghipour [42] provide a generalized validation framework that may be used for the CCSI efforts. Once experimental results are available from a bench-scale plant, currently under construction at the National Energy Technology Laboratory (NETL), validation of the regenerator CFD predictions will be performed. A description of the experimental setup can be found in Spenik [43]. One of the objectives of this work is to better understand the influence of operating conditions and guide future efforts toward the design of large-scale commercial devices.

3.1. Effect of side wall boundary conditions

The effect of employing different boundary conditions for the side walls also was examined. The free-slip, no-slip, and periodic boundary conditions were investigated. A baseline fill level of 20% sorbents by volume, particle size of 350, and gas velocity of $5v_{g,mf}$ was used. For the different boundary conditions, the solid fraction (ϕ_s) distribution, residence time distribution between two consecutive plates, and gas phase pressure drop were compared. The mean solid fraction variation with height is presented in Fig. 3. The ϕ_s values plotted in Fig. 3 were obtained by averaging over 10 s of steady-state simulation data recorded at intervals of 0.1 s. No significant difference was observed between the different boundary conditions. The coefficient of variation of the relative mean square error, CV(RMSE), was used to quantify the difference between the different solid fraction distributions shown in Fig. 3. Assuming the free-slip wall condition to be the baseline case, the CV(RMSE) error between the no-slip and free-slip cases was computed using:

$$CV(RMSE) = \frac{\sqrt{\sum (\phi_{s,ns} - \phi_{s,fs})^2}}{\sum \phi_{s,fs}}, \quad (11)$$

where the subscripts ns and fs refer to no-slip wall and free-slip wall, respectively. The same axial lengths in the bed region of the regenerator were used to compute the summations in the numerator and denominator of Eq. (11). The CV(RMSE) error between the no-slip and free-slip cases was found to be 0.0493, which shows that both boundary conditions yield almost the same results for the axial solid fraction distribution. The CV(RMSE) error value between the periodic case and free-slip wall case was also calculated in an analogous manner and found to be 0.1031—which is larger than the CV(RMSE) error between the no-slip and free-slip cases but can still be considered as being relatively small. The simulation with periodic boundary conditions also indicated a slightly lower height of the free surface. However, these differences between the periodic and free-slip wall cases are not

significant and are not expected to affect other measurements. The nature of the observed fluctuations is discussed in greater detail in Section 4.1.

A separate comparison of the residence time distribution and vertical pressure drop for the three different boundary conditions, not presented here, showed almost no differences (refer to Section 4.2 and Appendix A for a description of the residence time distribution calculation). As expected, the small difference in bed height had no influence on the residence time measurements. For the remainder of this work, the free-slip boundary condition was assumed for the side walls.

3.2. Effect of grid resolution

Agrawal et al. [27] show that the ratio of grid size to particle diameter should be $O(10)$ to obtain grid-independent results. Simulations of large devices at high resolutions are computationally prohibitive as the number of cells needed would be $O(10^6)$ for 2D and $O(10^9)$ for 3D simulations. Li et al. [44] state that grid-independent CFD simulations of industrial-scale devices are not possible using current computational resources. To provide a better perspective on the results reported in this work, the effect of grid resolution was investigated. For a particle size of 350 μm , superficial gas velocity of $5v_{g,mf}$, and sorbent fill level of 20%, simulations performed with 48×200 cells and 96×400 cells were compared. Fig. 4 illustrates that the solid fraction variation does not change appreciably when the resolution is doubled, and small differences are observed only in the dilute freeboard region immediately above the bed surface. The CV(RMSE) error between the solid fraction distributions of the two grid resolutions, computed using an expression analogous to Eq. (11), was found to have a relatively small value of 0.0756. This finding is consistent with the study by Chalermssinsuwan et al. [18], who perform simulations with varying grid sizes and find no significant variation in the solid fraction distribution. However, their claim of grid-independent results may be premature as microscopic flow features may continue to change with increasing grid resolution.

The residence time distribution $E(t)$ of sorbents between two consecutive porous plates is strongly governed by the cluster flow and

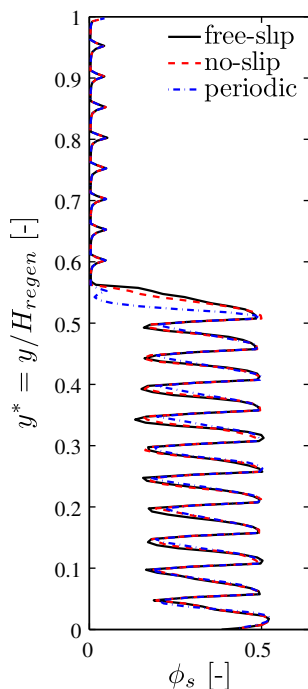


Fig. 3. Variation of mean solid fraction (ϕ_s) with dimensionless height for different boundary conditions.

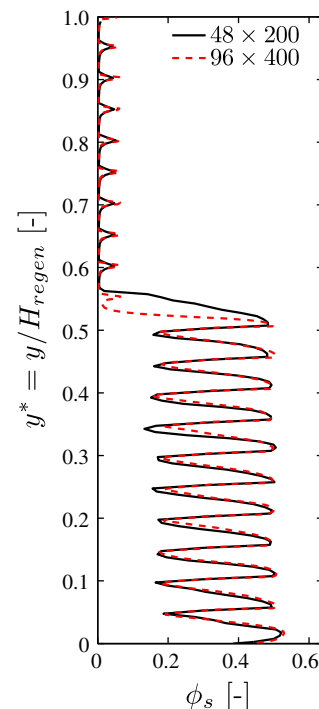


Fig. 4. Variation of mean solid fraction (ϕ_s) with dimensionless height at two different grid resolutions.

can be considered an indicator of microscopic flow. The calculation of $E(t)$ is detailed in Appendix A. A comparison of $E(t)$ in Fig. 5 for changing grid size shows some differences at smaller time values, i.e., in the neighborhood of the modes, but the absolute difference decreases farther away from the modes. The differences at smaller time values represent the behavior of clusters that reside in the section for short durations. However, the residence time distributions of sorbents that stay longer are captured fairly accurately even using a coarser grid. From these results, it may be summarized that the simulation at a lower resolution, i.e., with 48×200 cells, is able to capture the important macroscopic flow characteristics reasonably well. It should be noted that the measurements reported in this work may not yet be grid-independent—an estimated 1.6 million cells are required if a cell size ten times the particle diameter ($350 \mu\text{m}$) is to be simulated. Increasing the grid resolution does improve the microscopic results, but the improvement is not very large for macroscopic predictions such as axial solid fraction profile. Given the limitation on computing resources, a resolution of 48×200 cells is adopted for all simulations in the remainder of this work.

4. Results

4.1. Effect of sorbent mass holdup

The steady state volumetric holdup of sorbent particles (v_s) is a controllable parameter and may be used to prescribe the mean residence time of sorbents in the regenerator. The mean residence time $\bar{t}_{res}$ in the regenerator is given by:

$$\bar{t}_{res} = \frac{V_s}{Q_s}, \quad (12)$$

where V_s is the total volumetric sorbent holdup and Q_s is the volumetric flow rate. It is possible to control the holdup and flow rate in an actual device: one could initially keep the outlet blocked and after the desired holdup was achieved, adjust the outlet conditions to match the specified flow rate. Note that other parameters such as gas velocity, particle size, and internal plate geometry do not affect the mean residence time $\bar{t}_{res}$, but may affect the residence time distribution $E(t)$. The influence of varying sorbent holdup was investigated for a baseline particle diameter of 350 and superficial gas flow rate of $5v_{g,mf}$. Due to the use of equal sorbent mass flow rates at the top inlet and bottom outlet, the sorbent holdup in the regenerator was set by the initial conditions. In the

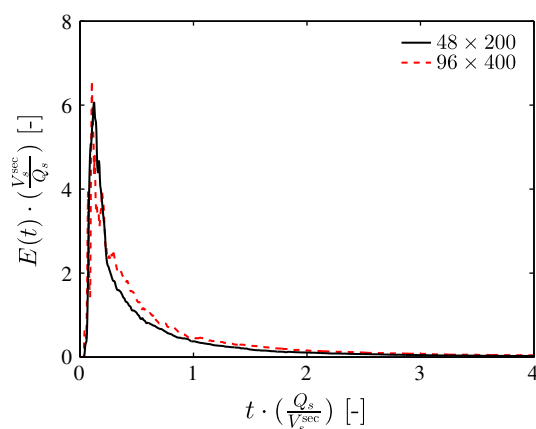


Fig. 5. Variation of residence time distribution ($E(t)$) between two consecutive plates at two different grid resolutions. The residence time distribution is made dimensionless using the ratio V_s^{sec}/Q_s , which represents the mean residence time between two consecutive plates (refer to Section 4.1). The quantity V_s^{sec} represents the volumetric sorbent holdup inside the section defined by the two plates and Q_s is the volumetric sorbent flow rate.

discussion that follows, the volumetric holdup v_s used to characterize the inventory of sorbents in the device is expressed as:

$$v_s = \frac{1}{V_{\text{regen}}} \int_{V_{\text{regen}}} \phi_s dV, \quad (13)$$

where ϕ_s is the solid fraction inside a small element, typically a cell, and dV is the volume of the element. The integral is computed over the total volume of the regenerator V_{regen} .

Fig. 6 presents the mean and standard deviation of the sorbent solid fraction as a function of height for varying volumetric holdup v_s . The value at each height is obtained by averaging over instantaneous solid fraction measurements at intervals of 0.1 s during the last 10 s of the simulation. Fig. 6 depicts the locations of the porous plates as gray bands. Similar to bubbling fluidized beds, two regions can be identified from Fig. 6(a) for each holdup: the lower bed region with a larger sorbent volume fraction and the upper freeboard region mostly occupied by the gas phase. Although the height of the interface between the two regions increases proportionately with holdup, the distribution of solids in either region does not change. In the upper freeboard region, only a small amount of sorbent particles is present at any given time. Moreover, these particles spend a very small time in the freeboard region as the bulk of total residence time is spent in the lower bed region. In an actual device, the height of the freeboard region should be minimized to reduce construction costs. Hence, the flow in the freeboard region is not of interest, and particle behavior in the lower bed region is analyzed in greater detail.

The $\phi_{s,\text{std-dev}}$ value in Fig. 6(b), the standard deviation of all ϕ_s measurements recorded during the last 10 s of the simulations, also does not show any change with holdup. Thus, the solid fraction distribution is found to be independent of the mass holdup, consistent with predictions from Ergun's equation (Eq. (2)). According to Ergun's equation, the solid fraction does not vary with total bed depth or location within the bed but depends on other parameters, such as superficial gas velocity and particle size.

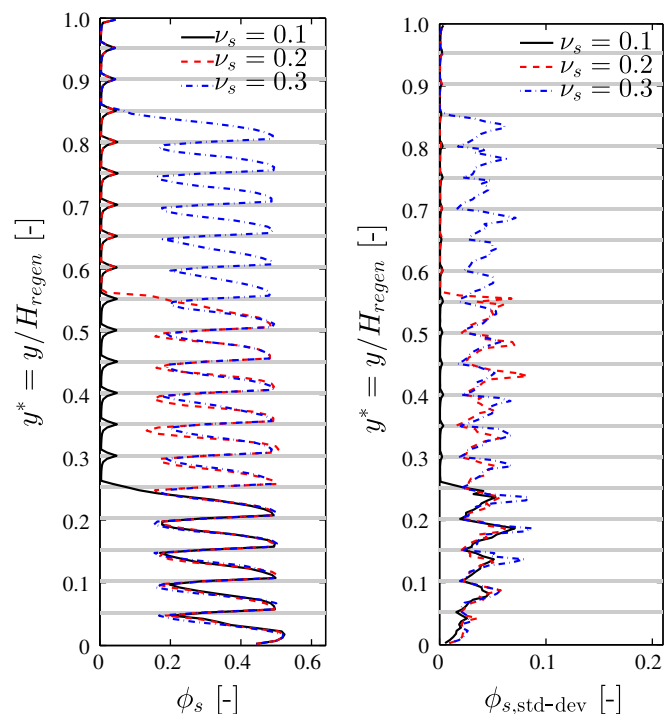


Fig. 6. Variation of (a) mean solid fraction ϕ_s and (b) standard deviation of solid fraction $\phi_{s,\text{std-dev}}$ with dimensionless height for varying solid holdup v_s . The standard deviation of solid fraction (b) represents fluctuations in local solid density due to formation and disintegration of particle clusters during the 10 s measurement interval.

For the cases reported in Fig. 6, increasing the holdup beyond a threshold value of 0.35 does not affect the solid fraction distribution in the lower region. Instead, the dilute upper region ceases to exist, and a dense-packed region starts to form at the top of the regenerator. To prevent the formation of dense packed regions near the roof of the regenerator, either the gas velocity should be less than the threshold value or the regenerator height can be increased.

The presence of perforated plates introduces significant fluctuations in solid fraction distributions when compared to fluidized beds without the plates. In systems without plates, such fluctuations are absent, and the solid fraction in the bed region is almost constant. For example, McKen and Pugsley [26], Xie et al. [45], and Li and Guenther [29] present typical solid fraction distributions that do not vary with height in the bed region. The fluctuations in the regenerator introduced by the porous plates, shown in Fig. 6(a), are qualitatively seen in all the cases discussed in this work, though the amplitude of the fluctuations may vary. The density of solids immediately above a porous plate is large, indicating that the plates support some part of the particle bed weight. The solid fraction decreases with increasing height until the lower extent of the plate above, beyond which the periodic fluctuations repeat. In the gas-dominated upper regions, similar periodic fluctuations are observed, but sorbent solid fraction values are much smaller.

Larger values of $\phi_{s, \text{std-dev}}$ in Fig. 6(b) correlated with smaller values of mean solid fraction in Fig. 6(a) and vice versa. At smaller solid fractions, fluidizing gas plays a more dominant role in formation and dissipation of dynamic particle clusters, which results in larger temporal fluctuations in solid fraction. Packed regions with larger solid fractions are less likely to exhibit chaotic flow behavior, resulting in smaller $\phi_{s, \text{std-dev}}$ values. The inclusion of porous plates significantly influences the fluidization characteristics of the bed by affecting the solid fraction holdup and dynamic clustering behavior. In the absence of porous plates, these fluctuations would be absent, and fluidization would be more homogeneous with height. The inhomogeneity in solids distribution introduced by the porous plates is likely to affect the chemical kinetics of sorbent regeneration in the device.

Fig. 7 presents the pressure distribution, made dimensionless using the outlet pressure P_{out} , as a function of dimensionless height. The pressure gradient is most significant in the regenerator's lower bed regions and is negligible in the upper freeboard region. Pressure fluctuations at any vertical location are negligible. The presence of the porous plates does not significantly affect the pressure drop. A linear fit using least squares shows that the pressure gradient does not change, even though the inlet pressure must be increased for larger holdups to maintain the prescribed gas flow rate.

The mean residence time through a section is governed by the volumetric flow rate and total volume or sorbents in the section (refer to

Eq. (12)), but the residence time distribution, in general, may be a function of other operating parameters and particle properties. The method used to determine the residence time distribution between two successive plates is described in Appendix A. A comparison of the residence time distributions between two adjacent plates for varying holdups (not included as a figure) indicated no significant differences, demonstrating that the holdup does not appear to influence the residence time distribution. A comparison of the residence time distribution over a fixed vertical length at different heights showed no notable differences (refer to Fig. A1 in Appendix A). From this observation, one can deduce that the residence time distribution between two plates in the bed region does not depend on the vertical location of the plates. The total residence time distribution of sorbents in the system E_{sys} may be obtained by repeated convolution of the residence time distributions through individual stages [46], given by:

$$E_{\text{sys}} = E_1 * E_2 * \dots * E_N, \quad (14)$$

where $E_1, E_2, \dots, E_N$ represent the residence time distributions in successive stages of the device. In the context of the regenerator, a stage is defined as the vertical space between two consecutive perforated plates. The convolution integral denoted by the asterisk is defined as:

$$E_i * E_j = \int_0^t E_i(\tau) E_j(t-\tau) d\tau = \int_0^t E_i(t-\tau) E_j(\tau) d\tau. \quad (15)$$

Varying the sorbent holdup would increase the number of plates submerged in the bed region and only influence the number of times, N , the convolution needs to be performed to obtain the total system residence time E_{sys} .

4.2. Effect of gas velocity and particle size

Particle size and gas velocity were systematically varied to study their effect on sorbent flow behavior. Because sorbent holdup does not influence behavior in the lower bed region of the current system, a constant holdup of 20% by volume was used for all simulations reported in this section. Six different particle sizes, ranging from 150 to 550, were simulated. For each particle size, the gas velocity was varied in the range of 1 to 10 times the respective minimum fluidization velocity $v_{g, \text{mf}}$. Gas velocities investigated for each particle size case are presented in Table 3.

The gas–solid vertical slip velocity $|v_g - v_s|$ and the particle diameter are expressed in terms of the dimensionless particle Reynolds number (Re_p) and Stokes number (St), given by:

$$Re_p = \frac{\rho_g d_p |v_g - v_s|}{\mu_g}, \quad (16)$$

$$St = \frac{(\rho_s - \rho_g) d_p^2 g}{18 \mu_g |v_g - v_s|}. \quad (17)$$

The gas and solid velocities in Eqs. (16) and (17) are computed by dividing the corresponding superficial velocities by the measured gas volume fraction and particle solid fraction, respectively.

Fig. 8 shows the mean solid fraction in the bed region as a function of the particle Reynolds number for varying particle diameter. Similar to an ordinary fluidized bed without perforated plates, the solid fraction was found to decrease with increasing gas fluidization velocity. The mean solid fraction values obtained from simulations were compared to predictions using the Ergun [23] and Yoon and Kunii [21] correlations. The Ergun [23] correlation (refer to Eq. (2)) was developed for a stationary fluidized bed and does not consider the velocity of solids. The Yoon and Kunii [21] correlation is a modification to the Ergun [23] equation for moving beds, where the superficial gas velocity $v_{g, \text{sup}}$ in Eq. (2) is replaced by the relative superficial velocity between the gas and solid phases, $(v_{g, \text{sup}} - v_{s, \text{sup}})$. Both models produce similar predictions for

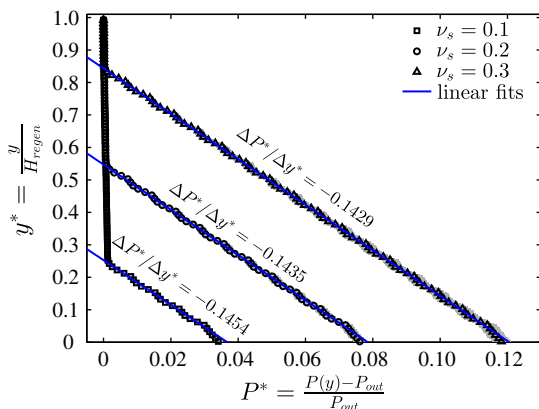


Fig. 7. Variation of dimensionless gas pressure (P^*) with dimensionless height (y^*) for varying solid mass holdup fraction ν_s .

Table 3

Values of the simulated particle size and superficial velocity of steam, expressed as a multiple of the minimum fluidization velocity $v_{g,mf}$. The $v_{g,mf}$ for each particle size is indicated in parenthesis below the particle size. The dimensionless Re_p and St values are also listed within parenthesis below the corresponding scaled gas velocities.

d_p ($v_{g,mf}$)	150 μm (0.0085 m/s)	200 μm (0.0151 m/s)	250 μm (0.0237 m/s)	350 μm (0.0464 m/s)	450 μm (0.0767 m/s)	550 μm (0.1146 m/s)
	1.0	1.0	1.0	1.0	1.0	1.0
	(0.23, 13.00)	(0.45, 15.66)	(0.79, 17.35)	(1.94, 19.46)	(3.91, 20.48)	(6.90, 21.19)
	1.7	1.7	1.7	1.7	1.6	1.6
	(0.30, 9.76)	(0.62, 11.32)	(1.13, 12.19)	(2.85, 13.22)	(5.54, 14.45)	(9.62, 15.19)
	2.7	2.7	2.7	2.7	2.5	2.5
$\frac{v_{g,sup}}{v_{g,mf}}$	(0.40, 7.33)	(0.85, 8.30)	(1.56, 8.80)	(4.01, 9.40)	(7.68, 10.42)	(13.07, 11.18)
(Re_p, St)	4.3	4.3	4.3	4.0	3.6	3.3
	(0.55, 5.37)	(1.19, 5.92)	(2.20, 6.23)	(5.33, 7.06)	(9.82, 8.15)	(15.70, 9.31)
	6.7	6.7	6.4	5.9	5.2	4.6
	(0.76, 3.92)	(1.65, 4.25)	(2.97, 4.63)	(7.00, 5.38)	(12.65, 6.33)	(19.72, 7.41)
	10.0	10.0	9.0	8.0	7.0	6.0
	(1.02, 2.92)	(2.22, 3.17)	(3.80, 3.61)	(8.62, 4.37)	(15.65, 5.12)	(24.31, 6.01)

cases where the solids superficial velocity is much smaller than the gas velocity, such as the larger particle size cases in Fig. 8. In general, both the Ergun [23] and Yoon and Kunii [21] empirical correlations underpredict the mean bed solid fraction in the regenerator. The agreement between simulation measurements and predictions appears to be better for the larger particle size cases. However, it should be noted that errors along the horizontal direction appear smaller on a logarithmic scale for larger values. A likely cause of the mismatch between measured and predicted solid fraction values is the introduction of porous plates in the domain, which affects the distribution of sorbent particles in the device. In ordinary fluidized beds without plates, the measured solid fraction values show good agreement with the established Ergun [23] or Yoon and Kunii [21] correlation. It should be noted that the data presented in Fig. 8 is corrected to account for the volume occupied by the porous plates.

Contours of the distribution of solids in the regenerator are presented for varying particle sizes in Fig. 9. A similar ratio of superficial gas velocity to minimum fluidization velocity is maintained for the different particle sizes. For smaller particle sizes, the sorbents are fluidized more uniformly, resulting in a more homogenized distribution of solids. For larger particle sizes, the bed has a more heterogeneous distribution of solids and demonstrates bubbling behavior. The smaller particle sizes can be classified as belonging to Geldart group A. Hence, they are fluidized easily and more uniformly [47]. However, the 550- μm -sized particles fall in Geldart group B, and bubbling behavior starts at incipient fluidization, resulting in the formation of larger, more heterogeneous solid clusters and gas bubbles. The sorbent regeneration rate is expected to differ for both smaller and larger particle size cases due to the change not only in particle size, but also in the formation of packed regions and inhomogeneous clusters of CO_2 -loaded sorbents. Situations resulting in

formation of large, dense-packed particle clusters are known to have smaller overall effective reaction rates compared to homogeneous gas–particle suspensions [28].

To identify operating conditions that result in a considerable fraction of the sorbents forming dense, close-packed regions, Fig. 10 plots the fraction of the bed volume with solid fraction greater than 58%. Henceforth, close packing shall refer to a solid fraction larger than 58%. The cases where more than 25% of the bed volume contains closely packed sorbents are further demarcated by a dotted line. The formation of dense-packed regions is more common for lower gas velocities and larger particle sizes. At lower gas velocities, Ergun's correlation (refer

$d_p = 150\mu\text{m}$	$d_p = 250\mu\text{m}$	$d_p = 550\mu\text{m}$
$\frac{v_{g,sup}}{v_{g,mf}} = 4.3$	$\frac{v_{g,sup}}{v_{g,mf}} = 4.3$	$\frac{v_{g,sup}}{v_{g,mf}} = 4.6$
$(Re_p = 0.55)$	$(Re_p = 2.20)$	$(Re_p = 19.72)$
$(St = 5.37)$	$(St = 6.23)$	$(St = 7.41)$

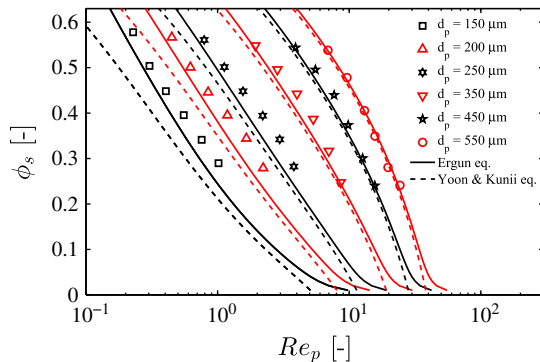


Fig. 8. The mean solid fraction ϕ_s in the active region of the bed for varying particle Reynolds number (Re_p) and Stokes number (St). The corresponding predictions obtained using the Ergun equation and the Yoon and Kunii equation are marked by solid and dotted lines, respectively. Note the logarithmic scale on the horizontal axis.

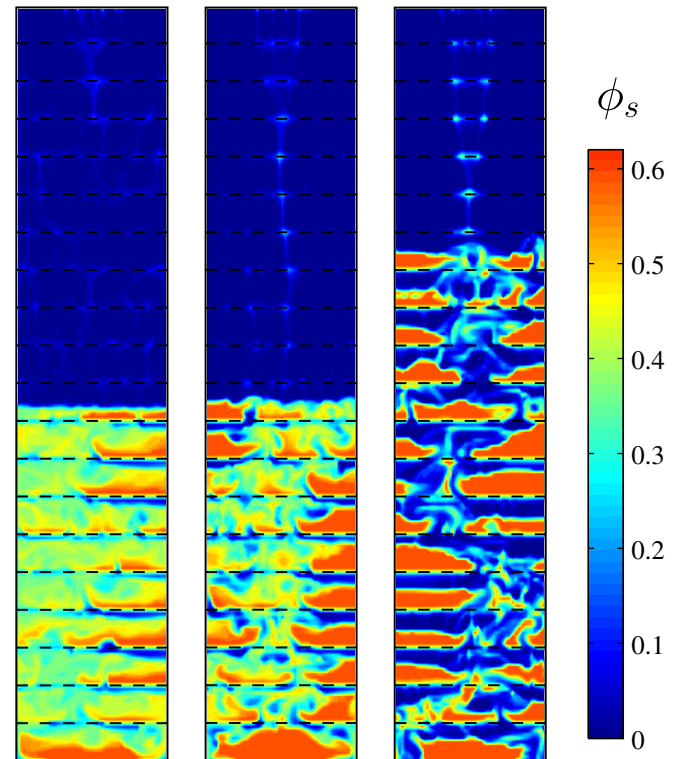


Fig. 9. Comparison of solid fraction distribution contours for varying particle sizes at 200 s. Similar values of the ratio $v_{g,sup}/v_{g,mf}$ are compared.

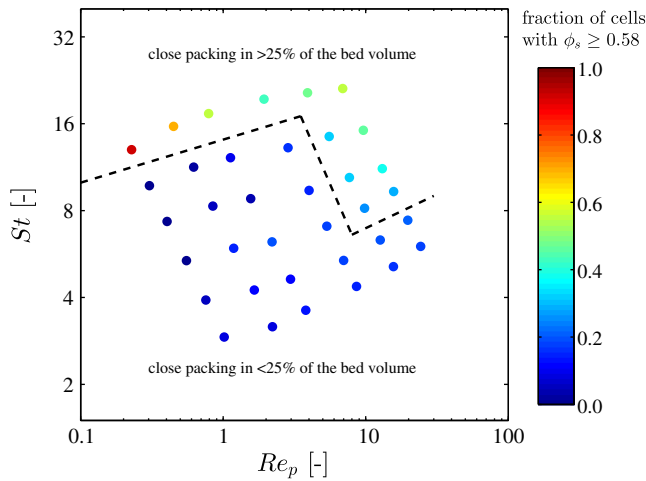


Fig. 10. Fraction of the bed volume with solid fraction ϕ_s larger than 0.58, representing close packing, for varying particle Reynolds number (Re_p) and Stokes number (St). In the cases above the dotted line, more than 25% of the bed volume—a considerable fraction—is found to have close solid packing.

to Eq. (2)) predicts a larger solid fraction, and formation of dense clusters is more likely. However, dense packing is also observed at higher gas velocities for larger sorbent diameters. This behavior corresponds to the onset of bubbling behavior as the powder transitions from Geldart group A to Geldart group B. For larger particles, the gas predominantly rises through channels formed between dense-packed solid clusters, resulting in potentially reduced reaction between the solid and gaseous phases. Reduced interaction between the CO_2 -loaded sorbent particles and superheated steam is expected to be unfavorable to sorbent regeneration.

The residence time distribution is also computed for the simulations reported in this section. The residence time between two adjacent porous plates is computed by injecting *in silico* tracer particles adjacent to a plate, then recording the concentration of the tracer particles at the vertical position located one plate spacing below the injection layer (refer to Appendix A). Based on the work by Gibilaro [48], the profile of tracer concentration with time was further processed to obtain the residence time distribution between the two plates. The location and time of tracer injection are found to have a negligible influence on the residence time distribution.

Depending on the particle size and gas velocity, the sorbent residence time distribution curves exhibit characteristics of both plug flow and well-mixed systems. Plug flow behavior is characterized by a narrow peak in the residence time distribution, suggesting a coherent downward sorbent movement with little axial mixing. The residence time distribution curves for the well-mixed cases span a wider time interval with a more gradual decay. Figs. A1 and A2 in Appendix A illustrate examples of residence time distributions for well-mixed flows.

The residence time distribution $E(t)$ obtained from the *in silico* tracer experiments was fitted using an analytical solution to the 1D advection–diffusion equation. The solution for open–open boundary conditions given by Gibilaro [48] is:

$$E(t) = \frac{L_{sec}}{t\sqrt{4\pi D_{yy}t}} \exp\left[-\frac{(L_{sec}-v_s t)^2}{4D_{yy}t}\right], \quad (18)$$

where L_{sec} is the length of the section under consideration, which currently is the vertical spacing between two plates. The solid velocity v_s is computed directly from the simulations. The axial diffusion coefficient for solids is denoted by D_{yy} and is a fitting parameter. The D_{yy} values obtained through least squares fitting is expressed as the dimensionless Peclet number, expressed as: $Pe = v_s L_{sec}/D_{yy}$. The Peclet number is a

ratio of advection rate to diffusion rate, which is helpful in discerning between plug flow and well-mixed flow behavior. Smaller Pe values are indicative of well-mixed flow, whereas larger values suggest plug flow.

Fig. 11 shows the Peclet number for varying Re_p and St as a color gradient. The boundary between plug flow and well-mixed flow in the present case is considered to be $Pe = 1$, although the transition value chosen by [46] is larger. The data points with a Peclet number smaller than one are demarcated from the larger Pe values by a dotted line. The confidence of these Pe measurements, i.e., the certainty of the fitted D_{yy} values, may be judged by the error between the measured and fitted (using Eq. (18)) residence time distributions. This error is qualitatively indicated using square symbols around the corresponding data points. Better fits (smaller squares) are generally obtained for the residence time distributions measured, particularly for the well-mixed flow cases. Larger errors are observed for some of the plug-flow cases, but closer inspection of the measured residence time distributions confirms plug-flow behavior. Therefore, Eq. (18) is a good model for the residence time distribution in a regenerator section and the uncertainties associated with most of the fitted D_{yy} values are small.

For larger particle sizes and smaller gas velocities, the Peclet number is found to be smaller than unity, suggesting a more well-mixed flow between the plates. For the smallest simulated particle size and larger gas velocities, the Peclet number is larger than unity, indicating behavior closer to plug flow. At larger gas velocities, the mean bed solid fraction is smaller as particles experience a greater degree of fluidization (refer to Fig. 8). To maintain the specified sorbent flow rate through the device, the mean solid velocity v_s increases at larger gas velocities. Hence, it can be argued that an increase in Pe values at higher gas velocities is due to larger advection rates, and not necessarily due to a decrease in diffusion rates. A closer examination of the absolute values of D_{yy} revealed that there is no systematic decrease in the diffusion coefficients with increasing gas velocity, confirming that the transition to plug flow is due to an increase in the advective transport rate of the sorbents.

It is still unknown if well-mixed flow or plug flow is more favorable to regenerator performance. If the sorbent particles at the regenerator inlet have a uniform CO_2 loading, plug flow is favored as it provides equal residence time for all sorbent particles and, hence, the same amount of regeneration. However, it is more likely for the sorbent particles to have a more heterogeneous distribution of adsorbed CO_2 . In such cases, some amount of axial diffusion is preferred to mitigate the

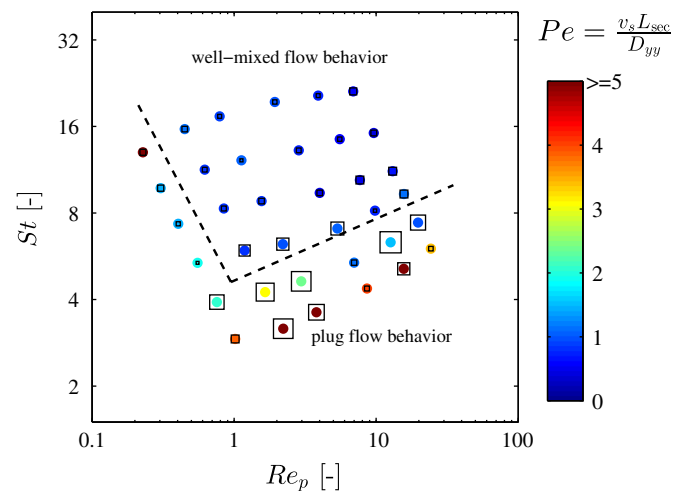


Fig. 11. The Peclet number (Pe) variation with particle Reynolds number (Re_p) and Stokes number (St). The well-mixed flow regime ($Pe < 1$) and the plug flow regime ($Pe > 1$) are roughly demarcated by the dotted lines. The size of the overlapping square symbols is proportional to the error between the measured and fitted residence time distribution, which indirectly quantifies the error in D_{yy} measurements.

fluctuations at the inlet and produce a more homogeneous quality of regenerated sorbent particles at the outlet. A large axial diffusion coefficient may not be ideal as it could lead to a large variation in the residence time of particles. Granular diffusion of sorbent particles inside clusters also is favorable for the operating cases where close-packed regions are more likely to occur. Local particle diffusion can help sorbent particles from the interior of the clusters reach the free surfaces and allow interaction with the superheated steam. The optimal value of the diffusion coefficient would be governed not only by the flow dynamics of the bed, but also by the chemical kinetics of CO₂ regeneration. Incorporation of chemical kinetics in the regenerator CFD model is the next step for this research.

5. Conclusions

A multiphase CFD model is developed for simulating the flow within the moving bed regenerator of a carbon capture unit. The influence of sorbent holdup, gas velocity, and particle size on flow behavior of the sorbent is systematically investigated. The inclusion of perforated plates, which subdivide the regenerator into a number of sections along the height, marks a significant departure in the current device's design compared to existing fluidized bed columns.

This work provides a better understanding of the flow inside the regenerator. The flow regime is governed primarily by the particle size and gas velocity (or alternatively, dimensionless Re_p and St values). Flow characteristics in the regenerator may be predicted using the regime maps presented in this study. For example, using smaller particles produces more homogeneous suspensions but limits the allowable flue gas flow rate. Larger gas flow rates may be attained using larger particles, but result in densely packed particle clusters. These dense clusters can potentially lead to reduced interaction between the sorbent and superheated steam. Achieving a balance between the flue gas flow rate and particle size will result in optimal regenerator performance.

The addition of porous plates introduces periodic fluctuations in the axial solid fraction distribution, which would otherwise be absent in systems without plates. An examination of the residence time distribution revealed well-mixed flow behavior for a wide range of operating conditions. The plug-flow assumption in previous models of carbon capture devices [1,2,14,11,8,9,7,10] needs to be re-examined based on the Peclet number, particularly for systems using solid sorbents.

Although the macroscopic solid fraction distribution is predicted reasonably well using larger grids, accuracy of the microscopic flow features (e.g., residence time distribution) improves upon grid refinement. For such large devices, grid-independent simulations are unlikely using present computational resources as even 2D simulations would require $O(10^6)$ cells. Alternate strategies, such as development of sub-grid 'filtered' models [35,36,28], must be investigated to accurately simulate larger devices. Future CCSI efforts will also focus on implementation of CO₂ desorption chemical kinetics in the CFD model. A set of suitable operating conditions identified from this work will form the basis of future models with chemical reactions.

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authors expressed herein do not necessarily state or reflect those of the United States Government or any agency thereof.

Notation

C_D	gas–solid drag coefficient, defined in Table 1
$CV(RMSE)$	coefficient of variation of the root mean square error
D_s	solid phase strain rate tensor, s^{-1}
D_{yy}	axial granular diffusion coefficient of sorbents, m^2/s
d_p	particle diameter, m
$d_{p,plate}$	assumed particle diameter of stationary plate phase, m
$E(t), E_j(t)$	residence time of (j th) section, s^{-1}
$E_{sys}(t)$	residence time through the system, s^{-1}
e_{pp}	particle–particle coefficient of restitution
e_{pw}	particle–wall coefficient of restitution
f_{gs}	specific gas–solid interaction force, N/m^3
g	specific body force, m/s^2
g	acceleration due to gravity, $9.81 m/s^2$
g_o	Lebowitz radial distribution function at contact [31]
H_{regen}	height of the regenerator, m
K_1, K_2, K_3, K_4	variables defined in Table 1
L_{sec}	section length between two porous plates, m
P	gas phase pressure, Pa
P_{out}	gas phase pressure at outlets, Pa
Pe	Peclet number
Q_s	sorbent volumetric flow rate, m^3/s
Re_p	particle Reynolds number, defined in Table 1
S	deviatoric solid strain rate, defined in Table 1, s^{-1}
St	particle Stokes number
t	time, s
$\bar{t}_{res}$	mean residence time, s
V_{regen}	internal volume of the regenerator, m^3
V_s^{sec}	volume of solids in a section between two plates, m^3
v_g, v_s	gas and solid phase velocities, m/s
$v_{g,mf}$	minimum fluidization velocity of gas, m/s
$v_{g,sup}$	superficial gas velocity, m/s

Greek letters

α	variable defined in Table 1
β_{gs}	gas–solid interphase drag coefficient, $kg/(m^3s)$
η	variable defined in Table 1
$\theta_{f,s}$	solid internal angle of friction
θ_s	granular temperature, m^2/s^2
μ, μ^*	variables defined in Table 1
μ_b, μ_s	bulk and shear solid viscosity, defined in Table 1
μ_g	gas phase viscosity
μ_{pw}	particle–wall friction coefficient
v_s	sorbent holdup volume fraction in the device
ρ_g, ρ_s	gas and solid phase densities, kg/m^3
σ_g, σ_s	gas and solid phase stress tensors, Pa
ϕ_g, ϕ_s	gas and solid phase volume fractions
ϕ_{plate}	porosity of stationary plate phase
$\phi_{s,max}$	maximum solid fraction at packing
$\phi_{s,mf}$	solid fraction at minimum fluidization
$\phi_{s,std} - dev$	standard deviation of ϕ_s fluctuations

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Appendix A. Calculation of residence time distribution for the sorbent

Although the mean residence time of sorbents is defined completely by the flow rate and volumetric holdup (see Eq. (12) in main text), the residence time distribution (RTD) may depend on the internal geometry, operating conditions, and sorbent properties. The RTD for a continuous flow system can be obtained either by injecting a tracer [46] or by solving the transport equation for local residence time [49]. To obtain the RTD for the full regenerator, a simulation with a duration of several times the mean residence time is required. Running a full-scale regenerator simulation for hundreds of seconds is computationally expensive. An alternate approach to obtain the total RTD is to divide the axial length of the regenerator into smaller sections and record the RTD through each section. Thus, the individual sectional RTDs obtained can be used to construct the RTD for the full regenerator via the convolution integral (refer to Eqs. (14) and (15) in main text and Levenspiel [46]).

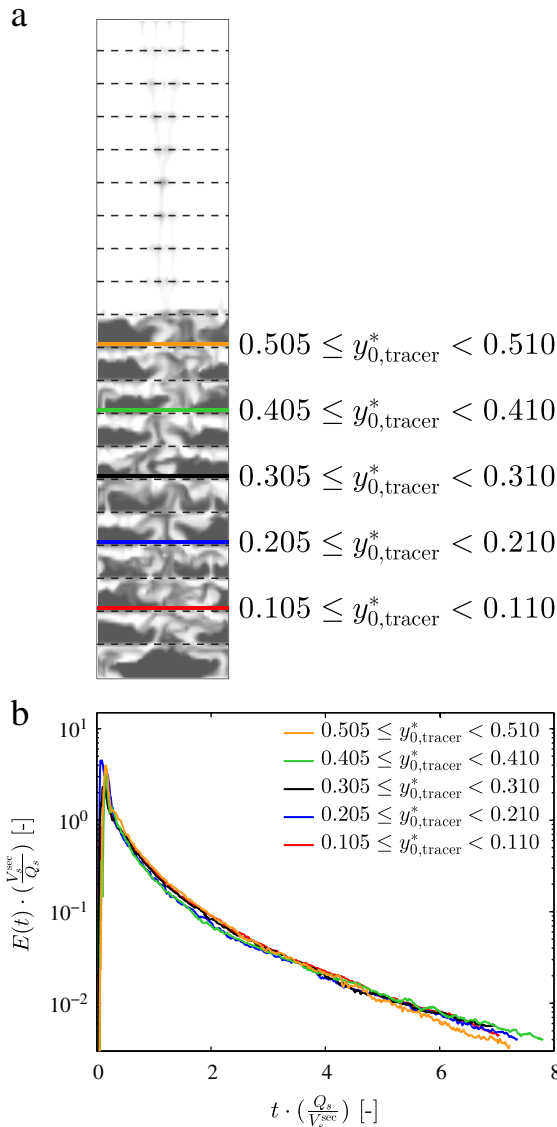


Fig. A1. The dimensionless residence time distribution $E(t)$ measured for different vertical sections. The initial locations of the tracer $y_{0,\text{tracer}}^*$ are marked in (a) and the residence time distributions plotted in (b) are measured over the section immediately below the initial tracer location.

For continuous flow systems with closed boundaries on both ends, calculation of the RTD using tracers is fairly straightforward. In a closed-boundary system, tracer particles can cross the boundaries only once. Hence the time between entry and exit is the exact residence time of a tracer particle. However, when a smaller axial section from the middle of the regenerator is considered, the boundaries of this section can no longer be considered as closed. Tracer particles are free to enter and exit the axial section at both ends and such a system is said to have open-boundary conditions. Gibilaro [48] addresses this issue and derives an equation for the RTD of open-boundary systems based on the 1D advection–diffusion equation, given by Eq. (18). The residence time relationship in Eq. (18) is found to be the response function to a unit impulse tracer input divided by the time after the input impulse was injected. Hence, impulse tracer experiments can be used to quantify the RTD for sections of the regenerator.

The bed region of the regenerator is divided into a number of equal-sized sections of height L_{sec} , where L_{sec} is the spacing between two consecutive porous plates. Once the flow attains steady state, the sorbent in a single horizontal layer of cells immediately above a plate is marked as tracer particles. Designating the particles of a single layer of cells in the simulation mimics the injection of an impulse input of tracers in an experiment without disturbing the flow. Locations of tracer “injection” in the regenerator at various heights are shown in Fig. A1(a). The concentration of the tracer particles is recorded at a downstream location situated a distance L_{sec} below the injection layer. The temporal variation of the tracer concentration at the downstream location is used to compute the RTD over the section based on Eq. (18) [48].

The RTD measured over different sections in the regenerator’s bed region is compared in Fig. A1(b). A sorbent holdup of 20% by volume, particle size of 350, and gas velocity of $5v_{g,\text{mf}}$ are used. The RTDs for the different sections show no perceivable differences—even for the sections near the upper freeboard region and bottom outlet. Section 1 describes that the solid fraction distribution is the same for different sections in the bed region. The match in RTD measurements further confirms that measurements made over one section of the bed region are representative of the flow in the entire bed region. Hence, in this work, RTD measurements over a single section in the bed region are reported and analyzed. It should be noted that the gas flow rate is uniform along the height of the regenerator in this model as the chemical kinetics of CO_2 desorption are not yet implemented.

The flow behavior in the bed region can be characterized as chaotic. Over a short time span, the flow is unsteady with rapid formation and dissociation of particle clusters. This unsteady and chaotic flow behavior

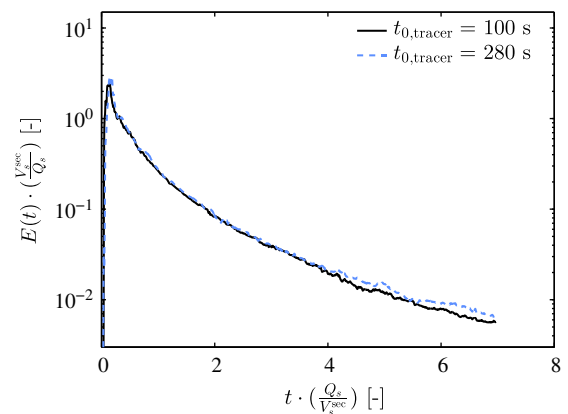


Fig. A2. The dimensionless residence time distribution for two sets of tracers injected at different time instants ($t_{0,\text{tracer}}$). Small differences in the two distributions suggest that dynamic clustering behavior in the chaotic flow may not be affecting RTD measurements significantly.

may affect the RTD if measured at different times. A more accurate RTD measurement may be obtained by calculating an ensemble average of RTD curves from multiple simulation replicates and multiple tracer injections at different times. To evaluate if such a computationally expensive approach is necessary, the RTDs for two sets of tracers injected at different times are presented in Fig. A2. The two curves show little difference, suggesting that chaotic flow does not significantly affect the RTD measurements. The small differences between the two RTD curves are most likely due to small differences in the unsteady bed behavior at different times. However, a single RTD measurement appears to be sufficient, and the small improvement achieved by calculating an ensemble average does not justify the increase in computational costs.

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